

Valid Inference: Possibilistic Inferential Models (with application in decision-making)

Shih-Ni Prim

Joint Work with Ryan Martin and Jonathan P. Williams

Seminar at Lawrence Livermore National Laboratory
August 26, 2026



WAKE FOREST
UNIVERSITY

Department of Statistical Sciences

NC STATE UNIVERSITY

Table of contents

- 1 Statistical Inference and the dichotomy of Frequentist vs Bayesian
- 2 Possibilistic Inferential Models
- 3 Decision making using IMs
- 4 Other extensions
- 5 Appendix

Section 1

Statistical Inference and the dichotomy of Frequentist vs Bayesian

Learning about the world

- We strive to learn about the world.
- What do we use to achieve this goal?
 - scientific objective
 - data
 - statistical model
- With data and a model, we can then do **inference**, prediction, decision making, etc.

Frequentist approach

- “Frequentist” methods pretty much mean “not Bayesian,” but many of them are likelihood based.
- Assuming the data X comes from a statistical model P_{Θ} , we want to know something (**inference** and **uncertainty quantification**) about the true parameter Θ .
- With observed data $X = x$, we can optimize the likelihood function $p_x(\theta)$ to find the maximum likelihood estimator $\hat{\theta}$, its sampling distribution, and confidence intervals.
- We can also study the asymptotic properties of how $\hat{\theta}$ converges to the true Θ .
- The sampling distributions and asymptotic properties provide error rate guarantees.

Frequentist approach, cont

- Using the above information, we can do inference about the parameter.
- However, with any hypothesis, we often conclude if it is rejected or not.
- We cannot quantify the confidence about a given hypothesis, for example,

$$\Theta > 2$$

given the observed data. That is, no probabilistic assessments (uncertainty quantification) of hypotheses are readily available.

Statisticians want numerical measures of the degree to which data support hypotheses. —Ian Hacking, *Logic of Statistical Inference*.

Bayesian methods

- Bayesian methods provide a convenient option for uncertainty quantification.
- Posterior distribution is derived from likelihood and prior:

$$q_x(\Theta) \propto p_x(\Theta)q(\Theta).$$

- A (precise) probability distribution is required for the prior.
- If nothing is known, a uniform prior is often used. But it encodes **indifference** instead of **ignorance**?
- Multiple “uninformative” priors can lead to different posteriors.

False confidence theorem

- High posterior probabilities can be assigned to false hypotheses. This happens, but we have to know and control the rate.
- In the Bayesian updating framework, compatible but not-really-true hypotheses can still get small posterior probabilities.
- Due to additivity, many such posterior probabilities can add up to not-so-small probabilities.
- This is the essential reason for false confidence.
- Other posterior-like probability frameworks—such as fiducial—also suffer for this problem.
- See Balch et al. (2019), Martin (2023a, 2024)
- We want both **reliable inference** and **uncertainty quantification**.

Section 2

Possibilistic Inferential Models

Methods outside the dichotomy

- Both frequentist and Bayesian methods use precise probabilities.
- Many others have been seeking alternatives outside the mainstream methods.
- Methods of imprecise probabilities:
 - Fuzzy set theory (Zadeh 1978)
 - Belief functions (Dempster 1966, Shafer 1976)
 - Generalized Bayes (Walley 1991)
 - Choice functions (Seidenfeld et al. 2010)
 - Possibility theory (Dubois & Prade 1988)

Possibilistic Inferential Models (IMs)¹

- Data-driven imprecise probabilities:
 - Given $X = x$ and an assumed model, for every hypothesis H , a pair of probabilities are assigned:

$$\sqcup_x(H) \text{ and } \sqcap_x(H)$$

- Necessity (lower probability) and possibility (upper probability)
 - Maxitivity instead of additivity
- Bayesian-like update without requiring a prior

...the most important unresolved problems in statistical inference is the use of Bayes theorem in the absence of prior information. —Brad Efron

¹Martin & Liu (2013), Martin (2026b)

Construct a possibilistic IM

- Relative likelihood:

$$R(x, \theta) = \frac{p_{\theta}(x)}{\sup_{\vartheta} p_{\vartheta}(x)}, \text{ with } \sup_{\theta} R(x, \theta) = 1.$$

- Possibilistic contour function:

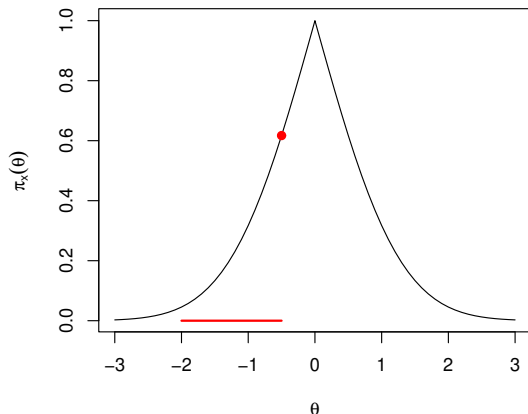
$$\pi_x(\theta) = P_{\theta}\{R(X, \theta) \leq R(x, \theta)\}.$$

- Then the corresponding possibility measure,² or upper probability, is defined as

$$\Pi_x(H) = \sup_{\theta \in H} \pi_x(\theta), \quad H \subseteq \mathbb{T}.$$

²Dubois & Prade (1988)

Possibility contour function



- Possibilistic IMs can **quantify (epistemic) uncertainty**.
- For $H : \theta \in [-2, -0.5]$, $\Pi_x(H) = \sup_{-2 \leq \theta \leq -0.5} \pi_x(\theta) \approx 0.617$.

Key properties

- The IM construction has the **validity** property:

$$\sup_{\Theta \in H} P_{\Theta} \{ \Pi_X(H) \leq \alpha \} \leq \alpha, \quad \alpha \in [0, 1]. \quad (1)$$

- **Error rate / reliability guarantee** (aleatory uncertainty quantification): *It's a small probability event that the IM assigns small upper probability to a true hypothesis.*
- The upper α set of the possibility contour

$$\mathcal{C}_{\alpha}(x) = \{ \theta \in \mathbb{T} : \pi_x(\theta) > \alpha \}, \alpha \in [0, 1]$$

is a $100(1 - \alpha)\%$ frequentist **confidence set**.

“Imprecision” in IMs

- A lower probability (necessity measure) also exists.

$$\sqcup_x(H) = 1 - \Pi_x(H^c).$$

- IMs are data-dependent and imprecise-probabilistic:

$$x \mapsto (\sqcup_x, \Pi_x)$$

- A credal set

$$\mathcal{C}(\Pi_x) = \{Q_x \in \text{probs}(\mathbb{T}) : Q_x(\cdot) \leq \Pi_x(\cdot)\} \quad (2)$$

characterizes the correspondence between IMs and precise probabilities.

Section 3

Decision making using IMs

Decision-making

- Let $(\theta, a) \mapsto \ell_a(\theta)$ denote a loss function that measures the loss incurred by choosing the action $a \in \mathbb{A}$ when the parameter is θ .
- If the true parameter is known, $a^* = \arg \min_{a \in \mathbb{A}} \ell_a(\Theta)$.
- When the true parameter is unknown, we need a model to quantify uncertainty about Θ and to find the expected loss.
- Decision-making is essentially a ranking strategy—picking the action that **minimizes the expected loss**.
- Two important goals of decision-making are finding the optimal action and reliable uncertainty quantification for risk assessment.

Bayesian (probabilistic) decision-making

- Posterior comes from likelihood and prior:

$$Q_x(d\theta) \propto p_\theta(x)Q(d\theta).$$

- Expected loss is integrated from the posterior distribution. For an action a and loss function $\ell_a(\theta)$,

$$Q_x \ell_a = \int \ell_a(\theta) Q_x(d\theta).$$

- The optimal action is usually the one minimizing the loss:

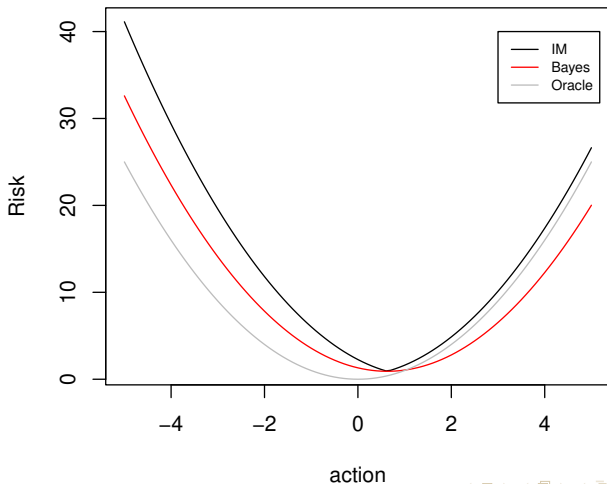
$$a^* = \arg \min_a Q_x \ell_a.$$

Example: Normal distribution

- Data: $X^n \stackrel{\text{iid}}{\sim} N(\Theta, 1)$, assume squared loss
- Bayesian model:
 - Prior: $\Theta \sim N(0, \tau^2)$
 - Posterior: $Q_x \sim N\left(\theta_{\text{MAP}}, \frac{\tau^2}{1+n\tau^2}\right)$, where $\theta_{\text{MAP}} = \frac{n\tau^2}{1+n\tau^2} \bar{X}_n$.
 - Expected Bayes loss: $(\theta_{\text{MAP}} - a)^2 + \frac{\tau^2}{1+n\tau^2}$.
- Expected IM loss: $(\bar{X}_n - a)^2 + 2|(\bar{X}_n - a)| m(p) + v(p)$, where $m(p) = \int |z| p(z) dz$, $v(p) = \text{var}(Z)$, $Z \sim N(0, \frac{1}{n})$.
- Oracle loss: $(\Theta - a)^2$

Comparing risk assessment

- One random draw of $n = 10$ from $\text{Normal}(0, 1)$ and then calculate losses from some actions $a \in [-5, 5]$.



Under-estimating risk is dangerous and can lead to systematic loss

On May 28, 2026, a New Glenn rocket operated by Blue Origin exploded during a static fire test.



Over-confidence in decision-making

- This is related to the false confidence theorem:
 - Bayes solutions, including those based on default-priors, and fiducial solutions are unreliable in that they can assign high posterior probability to hypotheses that are false. (Balch et al. 2019).
 - The core problem is **additivity**.

How do we fix the problem?

- Additivity is the root cause of false confidence. We need **non-additivity**, which means going outside the probabilistic framework.
- We need some sense of **reliability** guarantee, i.e. **aleatory** uncertainty quantification.
- We will use **possibilistic inferential models**,³ constructed in the possibilistic framework and not afflicted with false confidence.

³Martin & Liu (2013), Martin (2026*b*)

Use IMs in decision making⁵

- Expected IM loss is calculated by the Choquet integral:⁴

$$\begin{aligned}\Pi_x \ell_a &= \int_0^1 \left\{ \sup_{\theta: \pi_x(\theta) > s} \ell_a(\theta) \right\} ds \\ &= \sup \{ Q_x \ell_a : Q_x \in \mathcal{C}(\Pi_x) \}\end{aligned}$$

- This is the **upper expectation** and essentially the upper bound of the expected loss from the credal set (2).
- The optimal action is the minimizer of the Choquet integral.

$$\hat{a}(x) = \arg \min_a \Pi_x \ell_a.$$

- This follows the **minimax** principle!

⁴Troffaes & de Cooman (2014)

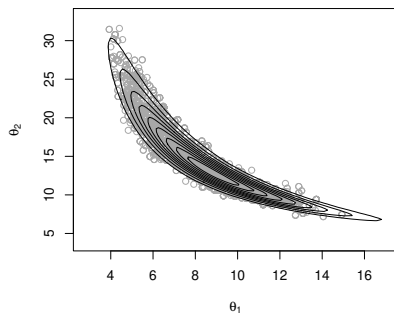
⁵Martin et al. (2026)

Application 1: Gamma model

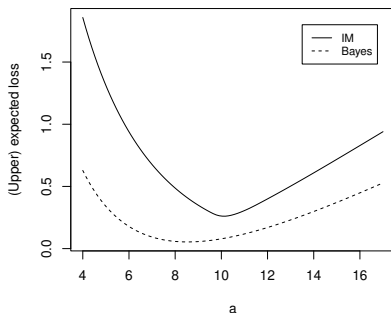
- The gamma distribution is commonly used as a model for time-to-event data, e.g., in survival or reliability analysis.
- $X = (X_1, \dots, X_n) \sim \text{Gamma}(\Theta_1, \Theta_2), \Theta_1 > 0, \Theta_2 > 0$.
- Consider the data in Example 3 of Fraser et al. (1997) on the survival time (in weeks) for $n = 20$ rats exposed to a certain level of radiation.
- For the decision-making problem, we focus here on estimating the shape using the loss

$$\ell_a(\theta) = (\log a - \log \theta_1)^2, \quad a > 0.$$

Application 1: Gamma model



(a) Contour function



(b) Risk assessments

Application 2: Random effects model

- A simplest random effects model is commonly written as

$$X_{ij} = \theta_0 + T_i + E_{ij}, \quad i = 1, \dots, m, \quad j = 1, \dots, n,$$

where $X = (X_{ij})$ is the observable data and T_i and E_{ij} are mutually independent random effects, with $E_{ij} \stackrel{\text{iid}}{\sim} N(0, \theta_1)$ and $T_i \stackrel{\text{iid}}{\sim} N(0, \theta_2)$.

- Then the three-dimensional parameter that indexes this model is $\theta = (\theta_0, \theta_1, \theta_2)$, where θ_0 is the overall mean and (θ_1, θ_2) are variance components associated with error/replication and treatment, respectively.
- The overall mean is not of primary interest, and it can be easily marginalized out, so the focus will be on the variance components, in particular, that for the treatment effect.

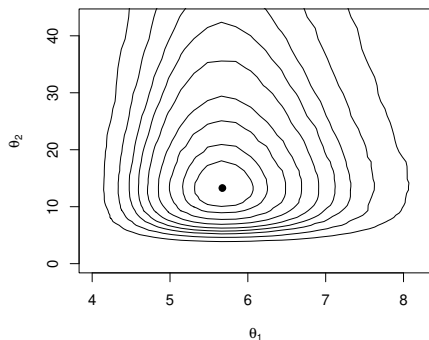
Application 2: Random effects model

- For estimation of the treatment variance component, Portnoy (1971) suggests the following variation on the usual squared-error loss function:

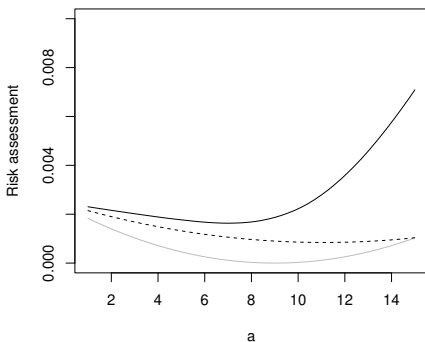
$$\ell_a(\theta) = \frac{(\theta_2 - a)^2}{(\theta_1 + n\theta_2)^2}, \quad a > 0. \quad (3)$$

- There are different ways to assess the risk associated with actions relative to this loss function. One is the oracle assessment, another is the Jeffreys-prior Bayes assessment as in (Tiao & Tan 1965), and the IM assessment proposed here.

Application 2: Random effects model



(c) Contour function



(d) Risk assessments

Panel (a) shows the possibilistic IM contour for the variance components (θ_1, θ_2) . Panel (b) compares the oracle (gray), Bayes (dashed), and IM (solid) risk.

Application 3: Laplace's rule of succession

- Sunrise problem; $X \sim \text{Bin}(n, \theta)$, $X = n$. $\text{MLE} = 1$.
- Laplace's rule of succession: $\hat{a}_{\text{LAPLACE}} = \frac{n+1}{n+2}$. ($\hat{a}_{\text{JEFFREYS}} = \frac{n+1/2}{n+1}$.)
- This agrees with the Bayes solution with uniform prior (or Jeffreys prior) and the squared loss $\ell_a(\theta) = (a - \theta)^2$.
- With IM constructions:

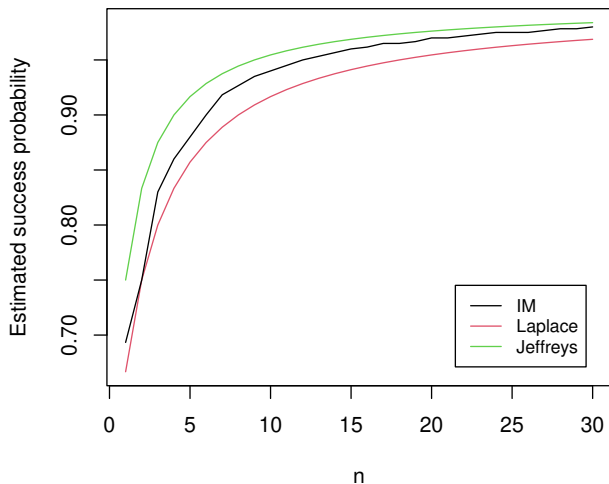
$$R(x, \theta) = \left(\frac{n\theta}{x} \right)^x \left(\frac{n(1-\theta)}{n-x} \right)^{n-x}.$$

$$\pi_n(\theta) = \sum_{k=0}^n \mathbf{1} \{R(k, \theta) \leq R(n, \theta)\} \cdot f_k(\theta),$$

where $f_k(\theta)$ is the probability mass function of a $X = k$ with the probability of success θ .

$$\hat{a}_{IM} = \arg \min_a \Pi_x \ell_a = \arg \min_a \int_0^1 \sup_{\theta: \pi_n(\theta) > s} (a - \theta)^2 ds.$$

Application 3: Laplace's rule of succession⁶



⁶arXiv:2608.11935

Shameless plug

International Journal of Approximate Reasoning 196 (2026) 109720



Contents lists available at ScienceDirect

International Journal of Approximate Reasoning

journal homepage: www.elsevier.com/locate/ijar



Decision-making with possibilistic inferential models^{*}

Ryan Martin , Shih-Ni Prim, Jonathan P. Williams

Department of Statistics, North Carolina State University, 2311 Stinson Drive, Raleigh, NC, 27695, USA



arXiv

Mathematics > Statistics Theory

(Submitted on 12 Aug 2026)

Induction and the rule of succession through a possibilistic inferential model lens

Ryan Martin, Shih-Ni Prim, Jonathan P. Williams

Induction is the process by which empirical evidence is transformed to knowledge. Hume famously argued—and Popper and others agree—that there can be no logical justification for induction. A weaker form of induction, due to Bayes, expresses the aforementioned knowledge in terms of probabilities, and we review some well-known and not-so-well-known criticisms of the Bayesian solution. We then investigate the relatively new possibilistic inferential model (IM) framework, showing that, in addition to the IM's strong, statistical reliability guarantees that it uniquely enjoys, it is safe from those criticisms that damage the Bayesian foundations. For illustration, we reconsider the classical sunrise problem and compare our proposed solution with Laplace's famous rule of succession.

Subjects: **Statistics Theory (math.ST)**; Other Statistics (stat.OT)

Cite as: [arXiv:2608.11935 \[math.ST\]](https://arxiv.org/abs/2608.11935)

(or [arXiv:2608.11935v1](https://arxiv.org/abs/2608.11935v1) [math.ST] for this version)

<https://doi.org/10.48550/arXiv.2608.11935> 

Section 4

Other extensions

Monte Carlo approximation

- Exact computation of contour function can be expensive, especially for high dimensions.
- For large enough sample size, we can use Monte Carlo and assume elliptical shape to draw posterior-like samples. These samples approximate the largest probability.
- The probability distribution is then transformed back to possibility contour. See Martin (2026a).

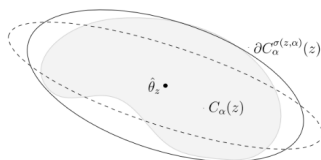
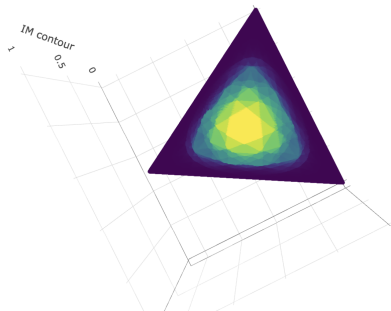
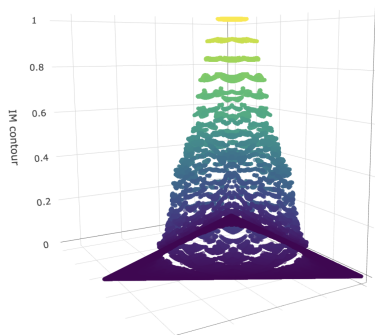


Figure 1: Approximating the IM contour's α -cut $C_\alpha(z)$ by an ellipse $C_\alpha^\sigma(z)$ with a “good” choice of σ (solid) and with a “bad” choice of σ (dashed).

Discrete distributions with continuous parameters

- Monte Carlo approximation is difficult for discrete (e.g. categorical) distributions.
- The approximate PDF/PMF is then a mixture.
- Ongoing work by Sahil Patel (NC State).⁷



⁷<https://sites.google.com/ncsu.edu/sahil-patel/>

Possibilistic IM prediction

- Define an “oracle” possibilistic predictor for a given θ :

$$\pi_x^{\text{OR}}(\tilde{x}|\Theta) = P_{\Theta}\{p_{\Theta}(\tilde{X}) \leq p_{\Theta}(\tilde{x})\}.$$

- Since the true Θ is unknown, average the oracle possibilistic predictors with respect to Π , using Choquet integration. This is the same strategy described in the decision-making context.

$$\pi_x^{\text{CH}}(\tilde{x}) = \int_0^1 \left\{ \sup_{\theta: \pi_x(\theta) > s} \pi_x^{\text{OR}}(\tilde{x}|\theta) \right\} ds.$$

- This could be too conservative, but we can combine computational strategy⁸ to increase efficiency.
- Ongoing work by James Robertson (NC State).

⁸Martin (2026a)

Other extensions

- Supervised learning (Cella & Martin 2022)
- Divide and conquer (Hector et al. 2025)
 - Validate the sampling distribution of the block summaries to obtain finite-sample validity guarantee.
- Data fusion (Cella 2026)
 - Combine possibility measures while preserving validity.
 - Useful for meta-analysis and data splitting.
- Distribution free inference (Cella 2024)
 - Given the observed data, infer a quantile or predict the next observation without specifying a parametric model.
- Structural learning (Martin et al. 2025)
- If there is any prior knowledge, we can encode partial priors. (Martin 2023b)

Summary

- Statistics aims to understand the world—“find the truth”—given the evidence (observable data).
- Both Frequentist and Bayesian approaches are not fully satisfactory.
- Looking beyond additivity and the probabilistic framework, possibilistic inferential models (IMs), situated in imprecise probabilities, provide a novel solutions for uncertainty quantification with reliability guarantee.
- Various extensions have been done. More are ongoing work.
- This is a powerful tool for problems with real-world impact and consequences—like those worked on in LLNL!
- Let me know if you are interested in applying IMs on a project!

Thank you!

Website: <https://snprim.github.io/>

Email: prims@wfu.edu

Section 5

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Section 6

Appendix

Theoretic properties: Finite-sample validity

- This is an extension from the validity property in (1).
- Let $\ell_a : \mathbb{T} \rightarrow [0, \infty)$ be a non-negative loss function, and define

$$\mathcal{R}(x, \Theta) = \inf_{a \in \mathbb{A}} \frac{\Pi_x \ell_a - \ell_a(\hat{\theta}_x)}{L_a(x, \Theta) - \ell_a(\hat{\theta}_x)} \geq 0, \quad (4)$$

where $L_a(x, \Theta)$ is

$$L_a(x, \Theta) = \sup\{\ell_a(\theta) : \pi_x(\theta) > \pi_x(\Theta)\}, \quad (5)$$

and $\hat{\theta}_x$ is the maximum likelihood estimator of Θ .

Finite-sample validity, cont

Theorem (Validity for Risk Assessment)

$$\sup_{\theta} P_{\theta} \{ \mathcal{R}(X, \theta) \leq \alpha \} \leq \alpha, \quad \text{for all } \alpha \in [0, 1]. \quad (6)$$

This also implies the simpler-but-weaker result

$$\sup_{\theta} P_{\theta} \left\{ \inf_{a \in \mathbb{A}} \frac{\Pi_X \ell_a}{L_a(X, \theta)} \leq \alpha \right\} \leq \alpha, \quad \text{for all } \alpha \in [0, 1].$$

Theoretic properties: Large sample efficiency

Theorem

Assume the regularity conditions as in Section 3.2 of Martin & Williams (2025) and summarized in Appendix A.2 of ?. For each $a \in \mathbb{A}$, let ℓ_a be a non-negative and continuous function with compact support. Then $\Pi_{X^n} \ell_a \rightarrow \ell_a(\Theta)$ in P_Θ -probability as $n \rightarrow \infty$, for each action $a \in \mathbb{A}$. Moreover, if the collection $\{\ell_a : a \in \mathbb{A}\}$ is uniformly bounded and equicontinuous, then the convergence is uniform in actions, i.e., $\sup_{a \in \mathbb{A}} |\Pi_{X^n} \ell_a - \ell_a(\Theta)| \rightarrow 0$ in P_Θ -probability as $n \rightarrow \infty$.

Appendix: Finite-sample validity

Proof.

Define the function

$$h_{x,a}(s) = \sup_{\theta: \pi_x(\theta) > s} \ell_a(\theta), \quad s \in [0, 1], \quad (7)$$

so that the Choquet integral $\Pi_x \ell_a$ is just a Riemann integral of $h_{x,a}$. It's clear that $s \mapsto h_{x,a}(s)$ is decreasing, which implies that, for any θ ,

$$\begin{aligned} \Pi_x \ell_a &= \int_0^1 h_{x,a}(s) ds \\ &= \left(\int_0^{\pi_x(\theta)} + \int_{\pi_x(\theta)}^1 \right) h_{x,a}(s) ds \\ &\geq \pi_x(\theta) h_{x,a}(\pi_x(\theta)) + \{1 - \pi_x(\theta)\} h_{x,a}(1), \end{aligned}$$

Appendix: Finite-sample validity

where $h_{x,a}(1) = \ell_a(\hat{\theta}_x)$. Since $L_a(x, \theta) = h_{x,a}(\pi_x(\theta))$, it follows that

$$\begin{aligned}\mathcal{R}(X, \Theta) \leq \alpha &\iff \Pi_X \ell_a - \ell_a(\hat{\theta}_X) \leq \alpha \{L_a(X, \Theta) - \ell_a(\hat{\theta}_X)\} \\ &\implies \pi_X(\Theta) \{L_a(X, \Theta) - \ell_a(\hat{\theta}_X)\} \leq \alpha \{L_a(X, \Theta) - \ell_a(\hat{\theta}_X)\} \\ &\iff \pi_X(\Theta) \leq \alpha.\end{aligned}$$

Validity implies the latter event has P_Θ -probability $\leq \alpha$, which proves (4). The simpler, second inequality (6) holds because we can ignore the non-negative “ $\{1 - \pi_x(\theta)\}h_{x,a}(1)$ ” in the lower bound for $\Pi_X \ell_a$ stated above. □

Appendix: Expected Bayes loss

- Data: $X^n \stackrel{\text{iid}}{\sim} N(\Theta, 1)$
- Prior: $\Theta \sim N(0, \tau^2)$
- Posterior: $Q_x \sim N\left(\theta_{\text{MAP}}, \frac{\tau^2}{1+n\tau^2}\right)$, where $\theta_{\text{MAP}} = \frac{n\tau^2}{1+n\tau^2}\bar{X}_n$.
- Expected Bayes loss:

$$\begin{aligned}
 Q_x \ell_a &= \int (\theta - a)^2 q_x(\theta) d\theta \\
 &= \int [(\theta - \theta_{\text{MAP}})^2 + (\theta_{\text{MAP}} - a)^2] d\theta \\
 &= (\theta_{\text{MAP}} - a)^2 + \frac{\tau^2}{1 + n\tau^2}
 \end{aligned}$$

Appendix: Expected IM loss

$$\begin{aligned}
 \pi_x(\theta) &= 2 \{1 - P(\sqrt{n}|\bar{X}_n - \theta|)\}. \\
 \bar{\Pi}_x \ell_a &= \int_0^1 \sup_{\theta: 2\{1 - P(\sqrt{n}|\bar{X}_n - \theta|)\} > s} (a - \theta)^2 ds \\
 &= \int_0^1 \max \left\{ \bar{X}_n - a \pm P^{-1}\left(1 - \frac{s}{2}\right) \right\}^2 ds \\
 &= \int_0^1 \left\{ |\bar{X}_n - a| + P^{-1}\left(1 - \frac{s}{2}\right) \right\}^2 ds \\
 &= \int_{-\infty}^{\infty} (|\bar{X}_n - a|^2 + |z|)^2 p(z) dz \\
 &= (\bar{X}_n - a)^2 + 2 |(\bar{X}_n - a)| m(p) + v(p),
 \end{aligned}$$

where $m(p) = \int |z|p(z)dz$, $v(p) = \text{var}(Z)$, $Z \sim N(0, \frac{1}{n})$, $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$.

Consider $a = \bar{X}_n$, $\Theta = 0$

- Expected Bayes loss:

$$\begin{aligned} E[Q_x \ell_a(\theta)] &= \frac{\tau^2}{1 + n\tau^2} + E[(\theta_{\text{MAP}} - \bar{X}_n)^2] \\ &= \frac{\tau^2}{1 + n\tau^2} + E\left[\left(\frac{n\tau^2}{1 + n\tau^2} - 1\right)^2 \bar{X}_n^2\right] \\ &= \underbrace{\left(\frac{n\tau^2}{1 + n\tau^2} + \frac{1}{(1 + n\tau^2)^2}\right)}_{=: h(n\tau^2)} \cdot \frac{1}{n} < \frac{1}{n}. \end{aligned}$$

- $h(n\tau^2) < 1$ for $n\tau^2 \in (0, \infty)$.
- Expected IM loss:

$$E[(\bar{X}_n - a)^2 + 2|(\bar{X}_n - a)|m(p) + v(p)] = v(p) = \frac{1}{n}.$$

- Oracle loss: $(0 - \bar{X}_n)^2 = \bar{X}_n^2$; $E(\bar{X}_n)^2 = \frac{1}{n}$.